

Suyash Bandi

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TECHNICAL SKILLS

C++, JavaScript, HTML, CSS, React.js, Tailwind, Node.js, Express.js, WebSockets, RESTful APIs, MongoDB, MySQL, Git, GitHub, Visual Studio Code, Python (ML scripting), Random Forest, Data Preprocessing

PROJECTS

GeoVista – Interactive Global Insights Platform — React, Vite, REST APIs, Netlify

- Developed a responsive React application displaying real-time geographic, political, and economic data for over 200 countries.
- Designed modular, reusable components with scalable architecture to support future feature expansion.
- Integrated REST APIs for real-time data fetching with error handling and caching mechanisms.
- Implemented advanced search functionality with autocomplete and dynamic filters to improve user experience.
- Deployed the application on Netlify ensuring automated continuous deployment and reliable global hosting.
- Optimized performance by implementing lazy loading and code splitting, improving load time by 30%.

CollaBoard – Real-Time Collaborative Whiteboard — Node.js, Express.js, Socket.IO, JWT

- Developed a real-time collaborative whiteboard application with multi-user drawing synchronization using WebSockets and Node.js.
- Implemented secure JWT-based authentication and role-based user permissions to manage access control.
- Built drawing logic using HTML5 Canvas and JavaScript for smooth and intuitive interaction.
- Designed REST APIs to manage sessions, boards, and user data effectively.
- Deployed backend on Heroku with monitoring tools to ensure uptime and performance.
- Handled data consistency and conflict resolution during concurrent drawing events.

Dissolved Oxygen Prediction Model — Random Forest, Machine Learning

- Developed a Random Forest machine learning model to predict dissolved oxygen levels in river water based on water quality parameters.
- Performed data preprocessing, feature selection, and normalization to improve model accuracy and generalization.
- Evaluated model performance using cross-validation and metrics such as R^2 and Mean Absolute Error.
- Created a CLI-based interface for easy input of water quality parameters and prediction of dissolved oxygen levels.
- Documented methodology, model insights, and usage instructions in a well-structured GitHub repository.
- Currently enhancing model with additional algorithms and external data sources to further improve robustness.

ACHIEVEMENTS & CO-CURRICULAR ACTIVITIES

Adobe India Hackathon – Round 1: Performed structured analysis of unstructured documents using heuristics and machine learning.

NPTEL Certification – Introduction to Algorithms and Analysis (IIT Kharagpur), Jul–Oct 2024, achieving a consolidated score of 78%.

Event Team Chief, VITronix Club, VIT Bhopal (2024-25): Organized and managed three major technical events with 300+ participants, ensuring successful execution and high engagement.

Solved over 200 problems on LeetCode to strengthen data structures and algorithms.

EDUCATION

Vellore Institute of Technology Bhopal

Bachelor of Technology in Computer Science and Engineering

Bhopal, MP

June 2022 – Present

St Arnolds Vijaynagar Indore

Senior Secondary (10th and 12th)

Indore, MP

April 2019 – July 2022